6 Network visualization

This chapter covers

- Creating adjacency matrices and arc diagrams
- Using the force-directed layout
- Representing directionality
- Adding and removing network nodes and edges

Network analysis and network visualization are more common now with the growth of online social networks like Twitter and Facebook, as well as social media and linked data in what was known as Web 2.0. Network visualizations like the kind you'll see in this chapter, some of which are shown in figure 6.1, are particularly interesting because they focus on how things are related. They represent systems more accurately than the traditional flat data seen in more common data visualizations.

This chapter focuses on representing networks, so it's important that you understand network terminology. In general, when dealing with networks you refer to the things being connected (like people) as *nodes* and the connections between them (such as being a friend on Facebook) as *edges* or *links*. You may hear nodes referred to as *vertices*, because that's where the edges join. Although it may seem useful to have a figure with nodes and edges labeled, one of the lessons from this chapter is that there is no one way to represent a network. Networks may also be referred to as